Rajalakshmi Engineering College

Department of Cybersecurity

CRYPTOGRAPHY LAB MANUAL

Name of the Student	ABISHEK S
Register Number	2116251901004
Subject	CRYPTOGRAPHY
Year	2nd YEAR

EXPERIMENT NO. 01

ENCRYPTION - CRYPTO 101

AIM: To learn the importance of cryptography and different encryption techniques and introduction to encryption.

TASK 2 :

- Are SSH keys protected with a passphrase or a password?
  → Passphrase

TASK 3 :

- What does SSH stand for?
  → Secure Shell
- How do web servers prove their identity?
  → Certificates
- What is the main set of standards you need to comply with if you store or process payment card details?
  → PCI-DSS

TASK 4 :

- What's 30%5?
  → 0
- What's 25%7?
  → 4
- What's 118613842%9091?
  → 3565

TASK 5 :

- Should you trust DES? Yea/Nay
  → Nay
- What was the result of the attempt to make DES more secure so that it could be used for longer?
  → Triple DES
- Is it okay to share your public key? Yea/Nay
  → Yea

TASK 6 :

- p= 4391, q= 6659, What is n?
  → 29239669

TASK 8 :

- What can you use to verify that a file has not been modified and is the authentic file as the author needed?
  → Digital Systems

TASK 9:

- What algorithm does the key use?
  → RSA
- Crack the password with the “John The Ripper” and “rockyou”, What’s the passphrase for key?
  → DELICIOUS

TASK 11 :

-
- You have the private key and a file encrypted with the public key, decrypting the file. What's the secret word?
  → Pineapple

Room completed: 100%

DO NOT encrypt passwords unless you're doing something like a password manager. Passwords should not be stored in plaintext, and you should use hashing to manage them safely.

Answer the questions below

What does SSH stand for?

Secure Shell ✓ Correct Answer

How do webservers prove their identity?

certificates ✓ Correct Answer

What is the main set of standards you need to comply with if you store or process payment card details?

PCI-DSS ✓ Correct Answer

Room completed: 100%

RSA and Elliptic Curve cryptography are based around different mathematically difficult (intractable) problems, which give them their strength. More about RSA later.

Answer the questions below

Should you trust DES? Yea/Nay

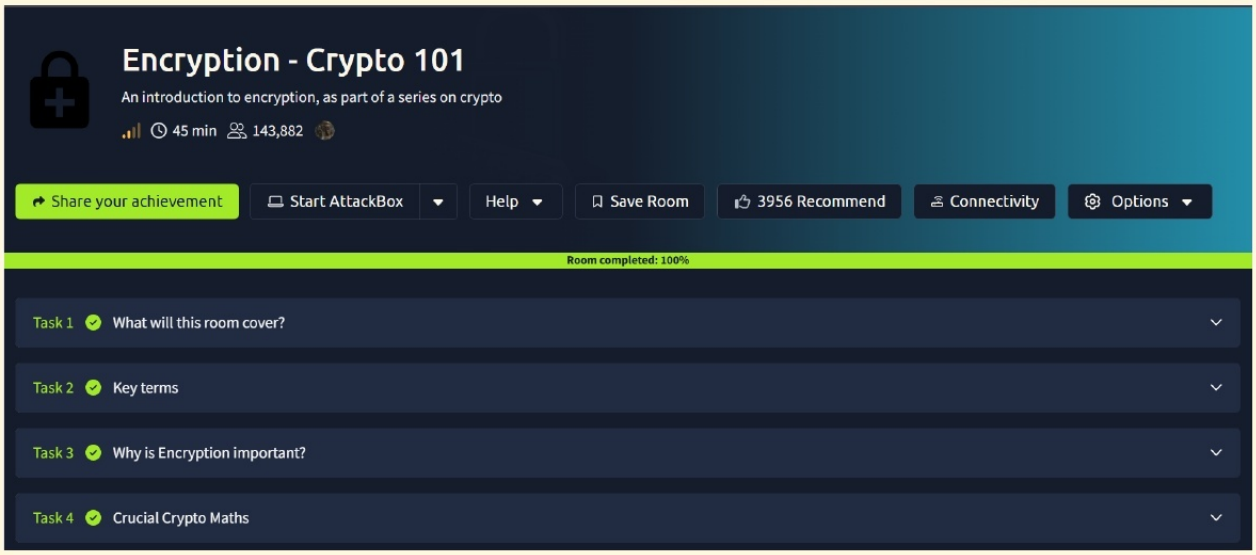
Nay ✓ Correct Answer

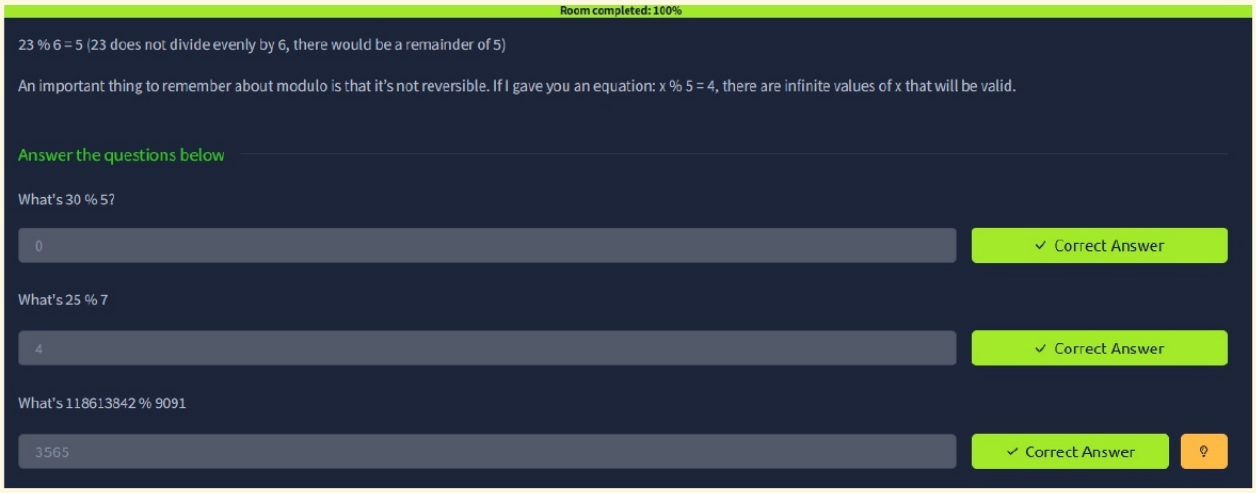
What was the result of the attempt to make DES more secure so that it could be used for longer?

Triple DES ✓ Correct Answer

Is it ok to share your public key? Yea/Nay

Yea ✓ Correct Answer





RESULT: Thus different encryption techniques and introduction to cryptography has been studied.

EXPERIMENT NO. 02

PASSIVE RECONNAISSANCE

AIM: To learn about the essential tools for passive reconnaissance, such as whois, nslookup, and dig.

TASK 2 :

- You visit the Facebook page of the target company, hoping to get some of their employee names. What kind of reconnaissance activity is this?
  → Passive reconnaissance
- You ping the IP address of the company webserver to check if ICMP traffic is blocked. What kind of reconnaissance activity is this?
  → Active reconnaissance
- You happen to meet the IT administrator of the target company at a party. You try to use social engineering to get more information about their systems and network infrastructure. What kind of reconnaissance activity is this?
  → Active reconnaissance

TASK 3 :

- When was TryHackMe.com registered?
  → 20180705
- What is the registrar of TryHackMe.com?
  → namecheap.com
- Which company is TryHackMe.com is using for name servers?
  → cloudflare.com

TASK 4 :

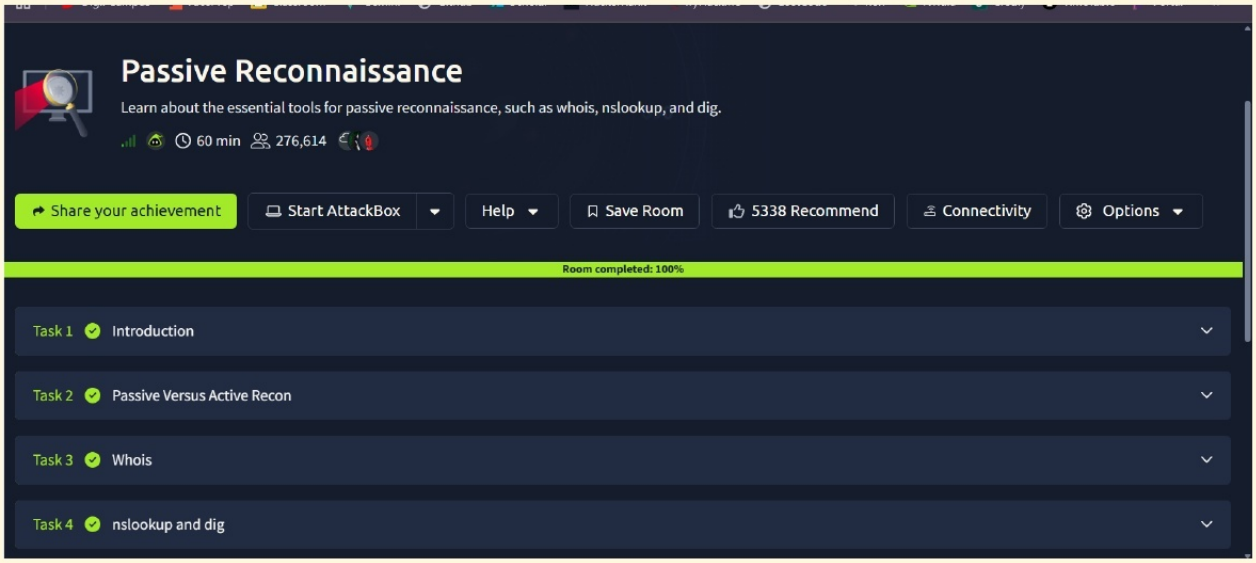
- Check the TXT records of thmlabs.com. What is the flag there?
  → THM{a5b83929888ed36acb0272971e438d78}

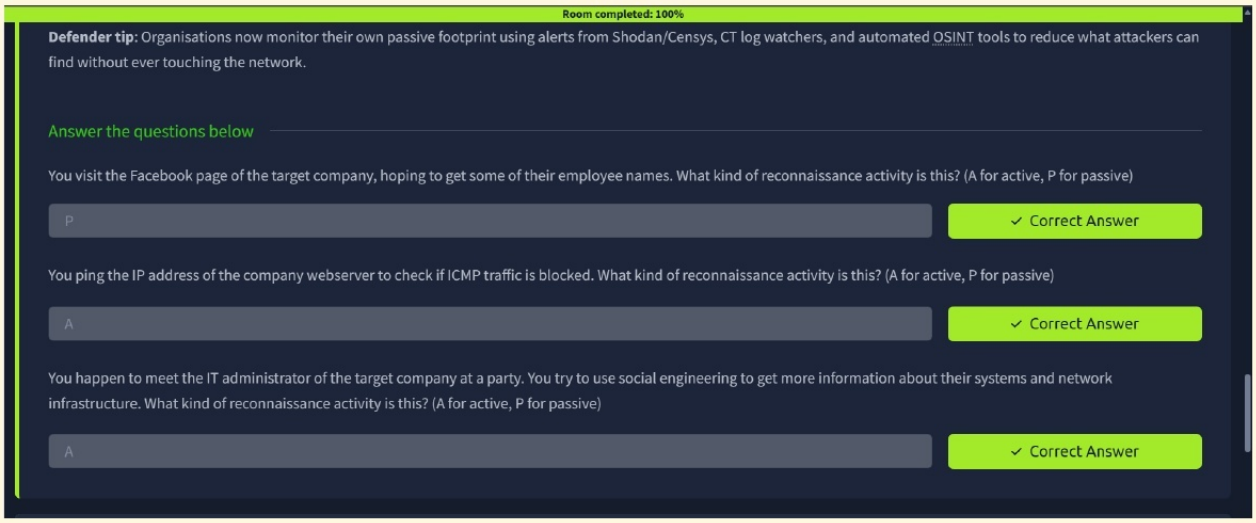
TASK 5 :

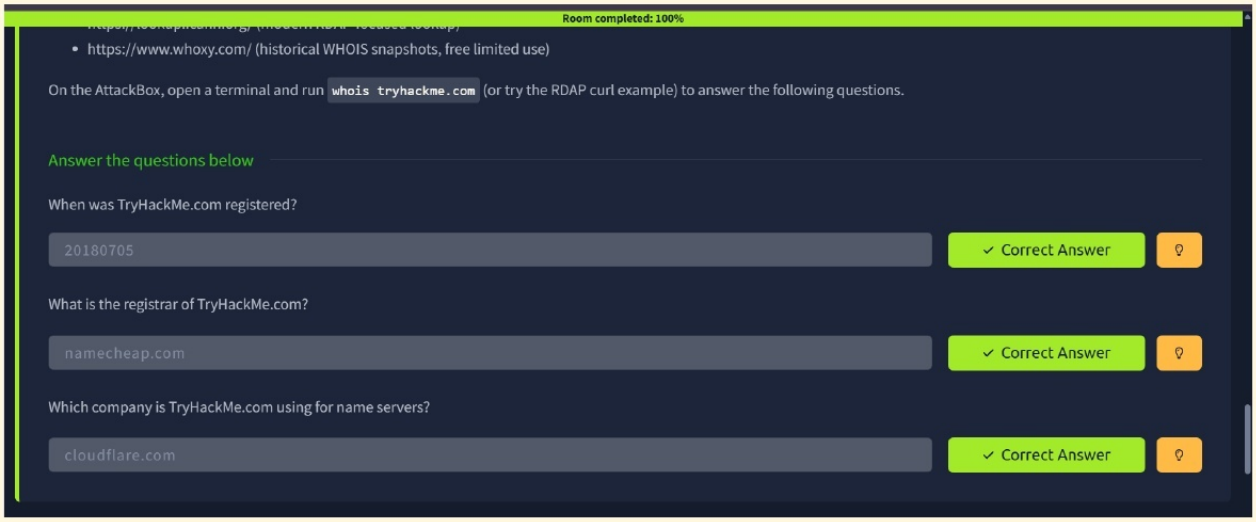
- Lookup tryhackme.com on DNSDumpster. Under **Services / Banners**, which one has the highest count?
  → cloudflare

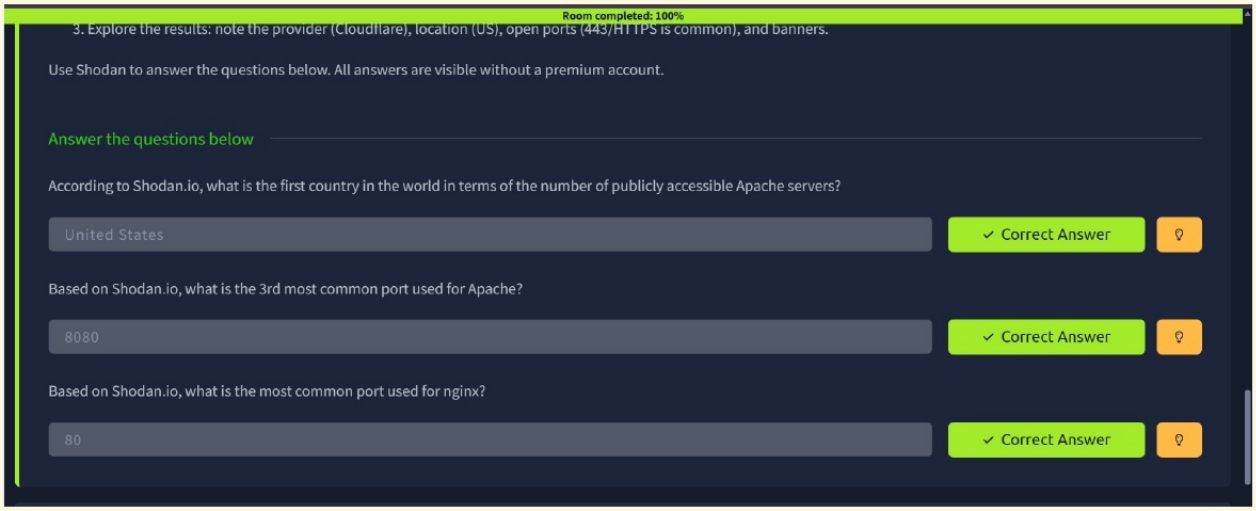
TASK 6 :

- According to Shodan.io, what is the first country in the world in terms of the number of publicly accessible Apache servers?
  → United States
- Based on Shodan.io, what is the 3rd most common port used for Apache?
  → 8080
- Based on Shodan.io, what is the most common port used for nginx?
  → 80









RESULT: Thus the essential tools required for passive reconnaissance have been studied.

EXPERIMENT NO. 03

ACTIVE RECONNAISSANCE

AIM: To learn how to use simple tools such as traceroute, ping, telnet, and a web browser to gather information.

TASK 2 :

- Browse to the following website (opens in new tab) and ensure that you have opened your Developer Tools on AttackBox Firefox, or the browser on your computer. Using the Developer Tools, figure out the total number of questions.
  → 8

TASK 3 :

- Which option would you use to set the size of the data carried by the ICMP echo request?
  → -s
- What is the size of the ICMP header in bytes?
  → 8
- Does MS Windows Firewall block ping by default? (Y/N)
  →Y
- Deploy the VM for this task and using the AttackBox terminal, issue the command `ping -c 10 MACHINE_IP`. How many ping replies did you get back?
  → 10

TASK 4 :

- In Traceroute A, what is the IP address of the last router/hop before reaching tryhackme.com?
  → 100.92.9.83

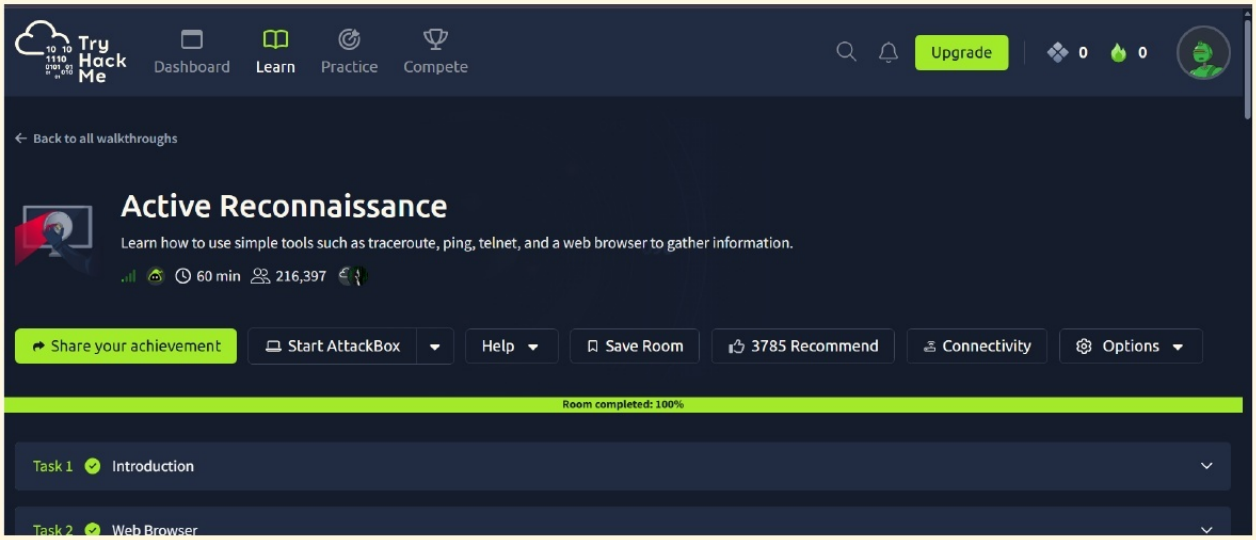
- In Traceroute B, what is the IP address of the last router/hop before reaching tryhackme.com?
  → 99.83.89.19
- In Traceroute B, how many routers are between the two systems?
  → 25

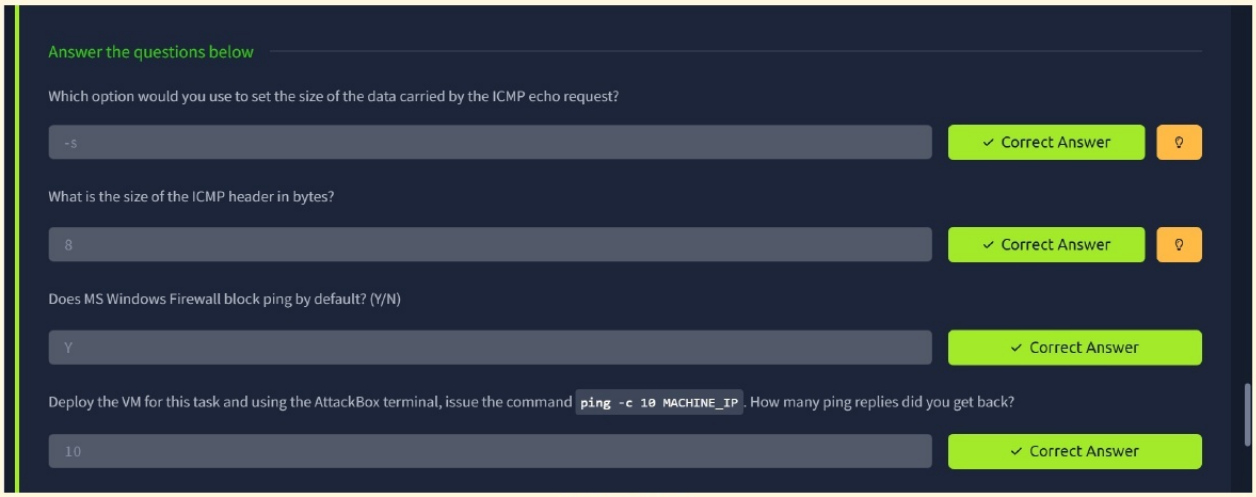
TASK 5 :

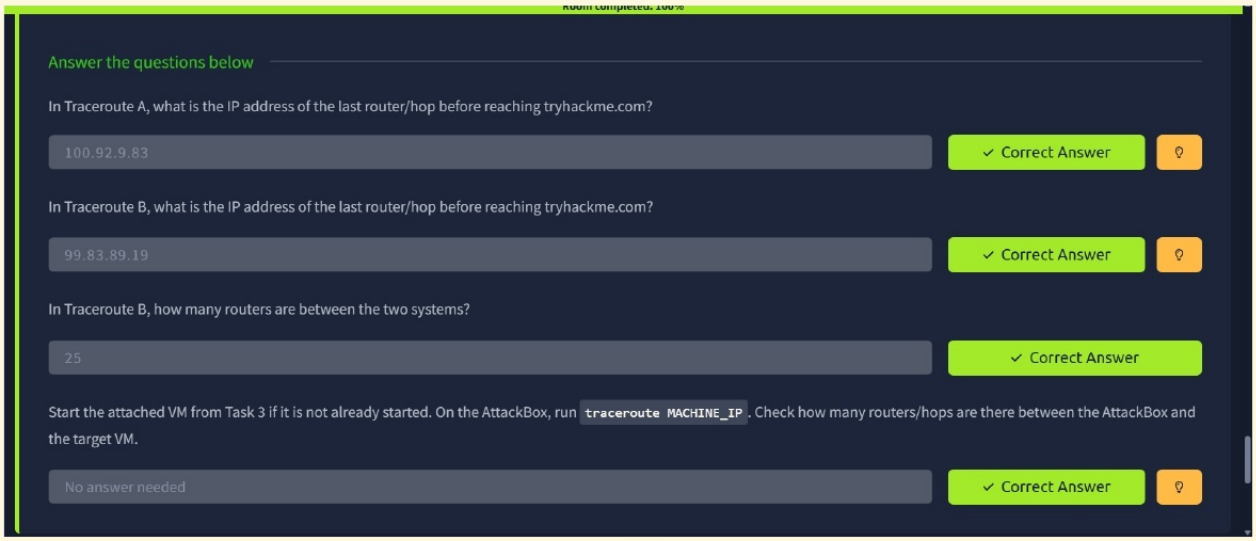
- Start the attached VM from Task 3 if it is not already started. On the AttackBox, open the terminal and use the telnet client to connect to the VM on port 80. What is the name of the running server?
  → Apache
- What is the version of the running server (on port 80 of the VM)?
  → 2.4.61

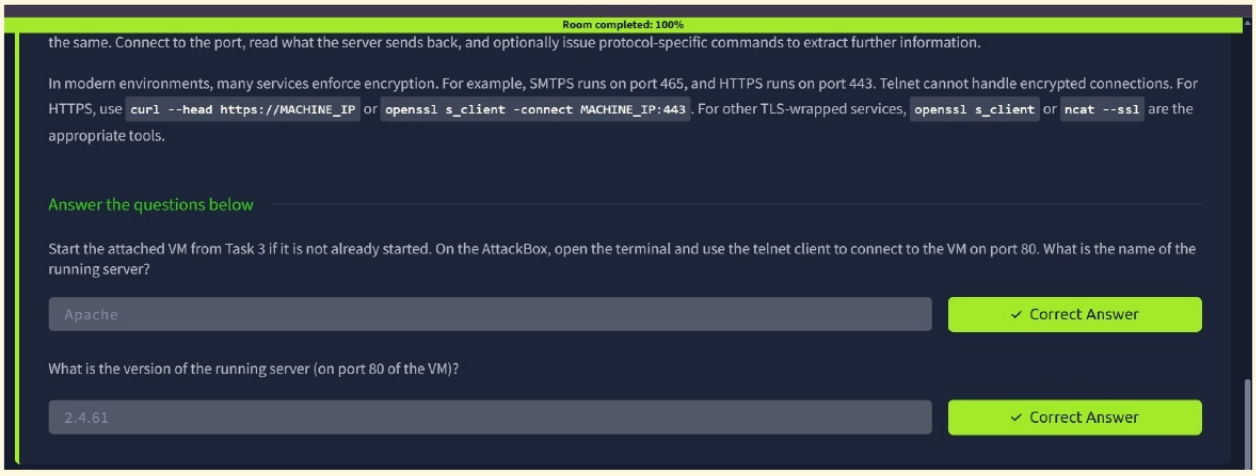
TASK 6 :

- Start the VM and open the AttackBox. Once the AttackBox loads, use Netcat to connect to the VM port 21. What is the version of the running server?
  → 0.17









RESULT: Thus the essential tools required for active reconnaissance have been studied.

EXPERIMENT NO. 04

SQL INJECTION LAB

AIM: To Understand how SQL injection attacks work and how to exploit this vulnerability.

TASK 2 :

- What is the flag for SQL Injection 1: Input Box Non-String?
  → THM{dccea429d73d4a6b4f117ac64724f460}
- What is the flag for SQL Injection 2: Input Box String?
  → THM{356e9de6016b9ac34e02df99a5f755ba}
- What is the flag for SQL Injection 3: URL Injection?
  →THM{645eab5d34f81981f5705de54e8a9c36}
- What is the flag for SQL Injection 4: POST Injection?
  → THM{727334fd0f0ea1b836a8d443f09dc8eb}

TASK 3 :

- What is the flag for SQL Injection 5: UPDATE Statement?
  → THM{b3a540515dbd9847c29cffa1bef1edfb}

TASK 4 :

- What is the flag for this challenge?
  → THM{f35f47dcd9d596f0d3860d14cd4c68ec}

TASK 5 :

- What is the flag for this challenge?
  →THM{fb381dfee71ef9c31b93625ad540c9fa}

TASK 6 :

- What is the flag for this challenge?
  → THM{f1f4e0757a09a0b87eeb2f33bca6a5cb}

TASK 7 :

- What is the flag for this challenge?
  → THM{4644c7e157fd5498e7e4026c89650814}

TASK 8 :

- What is the flag for this challenge?
  → THM{cd5c4f197d708fda06979f13d8081013}

TASK 9 :

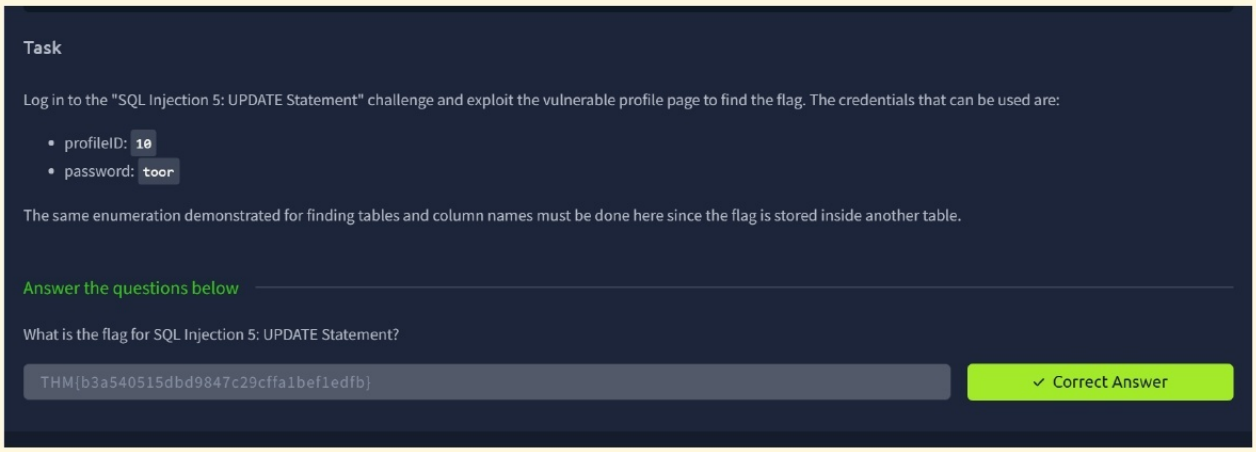
- What is the flag for this challenge?
  → THM{27f8f7ce3c05ca8d6553bc5948a89210}

TASK 10 :

- What is the flag for this challenge?
  → THM{183526c1843c09809695a9979a672f09}


SQL Injection Lab completed!
sufriyahussain.r completed another room on their cyber security journey.
Completed tasks
10
Points earned
96
Streak
1
Share on social media
Share your win before starting the next room
LinkedIn
WhatsApp
Telegram
Twitter / X
Facebook
Reddit


Answer the questions below
What is the flag for SQL Injection 1: Input Box Non-String?
THM{dccea429d73d4a6b4f117ac64724f460}
Correct Answer
What is the flag for SQL Injection 2: Input Box String?
THM{356e9de6016b9ac34e02df99a5f755ba}
Correct Answer
What is the flag for SQL Injection 3: URL Injection?
THM{645eab5d34f81981f5705de54e8a9c36}
Correct Answer
What is the flag for SQL Injection 4: POST Injection?
THM{727334fd0f0ea1b836a8d443f09dc8eb}
Correct Answer


Task
Log in to the "SQL Injection 5: UPDATE Statement" challenge and exploit the vulnerable profile page to find the flag. The credentials that can be used are:
profileID: 10
password: toor
The same enumeration demonstrated for finding tables and column names must be done here since the flag is stored inside another table.
Answer the questions below
What is the flag for SQL Injection 5: UPDATE Statement?
THM{b3a540515dbd9847c29cffa1bef1edfb}
Correct Answer

Task

Exploit the vulnerable login form and retrieve the flag.

Answer the questions below

What is the flag for this challenge?

THM{f1f4e0757a09a0b87eeb2f33bca6a5cb}

✓ Correct Answer

Task

Bypass the login and retrieve the flag.

Answer the questions below

What is the flag for this challenge?

THM{f35f47dcd9d596f0d3860d14cd4c68ec}

✓ Correct Answer

Task

Exploit the vulnerable login form and retrieve the flag.

Answer the questions below

What is the flag for this challenge?

THM{fb381dfee71ef9c31b93625ad540c9fa}

✓ Correct Answer

Task

Exploit the vulnerable function and retrieve the flag.

Answer the questions below

What is the flag for this challenge?

THM{4644c7e157fd5498e7e4026c89650814}

✓ Correct Answer

Task

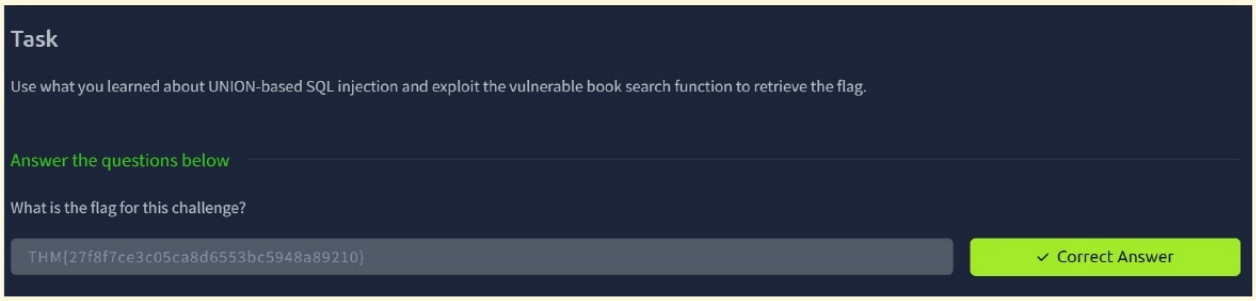
Create a new user and exploit the vulnerability in the update password function to access the admin account to get the flag.

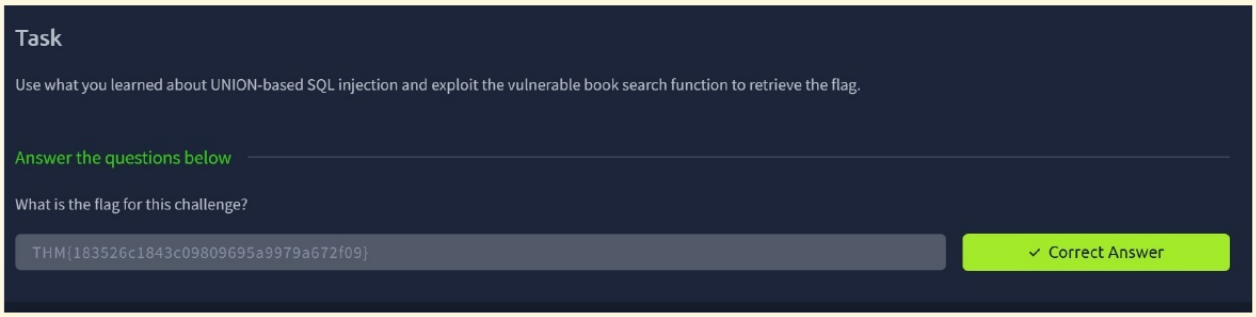
Answer the questions below

What is the flag for this challenge?

THM{cd5c4f197d708fda06979f13d8081013}

✓ Correct Answer





RESULT: Thus we understand how SQL injection attacks work and how to exploit this vulnerability.

EXPERIMENT NO. 05

SNORT IDS

AIM: To Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.

TASK 2 :

- Navigate to the Task-Exercises folder and run the command "./.easy.sh" and write the output
  → Too Easy!

TASK 3 :

- Which IDS or IPS type can help you stop the threats on a local machine?
  → HIPS
- Which IDS or IPS type can help you detect threats on a local network?
  → NIDS
- Which IDS or IPS type can help you detect the threats on a local machine?
  →HIDS
- Which IDS or IPS type can help you stop the threats on a local network?
  → NIPS
- Which described solution works by detecting anomalies in the network?
  → NBA
- According to the official description of the snort, what kind of NIPS is it?
  → full-blown
- NBA training period is also known as ...
  → baselining

TASK 4 :

- Run the Snort instance and check the build number.
  → 149
- Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.
  → 4151
- Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.
  → 1

TASK 6 :

- Now, you should have the logs in the current directory. Navigate to folder "145.254.160.237". What is the source port used to connect port 53?
  → 3009
- Read the snort.log file with Snort; what is the IP ID of the 10th packet?
  → 49313
- Read the "**snort.log.1640048004"** file with Snort; what is the referer of the 4th packet?
  → http://www.ethereal.com/development.html
- Read the "**snort.log.1640048004"** file with Snort; what is the Ack number of the 8th packet?
  → 0x38AFFFF3
- Read the "**snort.log.1640048004"** file with Snort; what is the number of the **"TCP port 80"** packets?
  → 41

TASK 7 :

- What is the number of the detected HTTP GET methods?
  → 2

TASK 8 :

- What is the number of the generated alerts?
  → 170
- Keep reading the output. How many TCP Segments are Queued?
  → 18

- Keep reading the output.How many "HTTP response headers" were extracted?
  → 3
- What is the number of the generated alerts?
  → 68
- What is the number of the generated alerts?
  → 340
- Keep reading the output. What is the number of the detected TCP packets?
  → 82
- What is the number of the generated alerts?
  → 1020

TASK 9 :

- Use "task9.pcap". Write a rule to filter IP ID "35369" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l . -r task9.pcap"
  → TIMESTAMP REQUEST
- Clear the previous alert file and comment out the old rules. Create a rule to filter packets with **Syn** flag and run it against the given pcap file. What is the number of detected packets?
  → 1
- Clear the previous alert file and comment out the old rules. Write a rule to filter packets with **Push-Ack** flags and run it against the given pcap file. What is the number of detected packets?
  → 216
- Clear the previous alert file and comment out the old rules. Create a rule to filter **UDP** packets with the same source and destination IP and run it against the given pcap file. What is the number of packets that show the same source and destination address?
  → 7
- Case Example - An analyst modified an existing rule successfully. Which rule option must the analyst change after the implementation?
  → rev



Snort completed!

sufriyahussain.r completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
11	232	2

Room completed: 100%

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS ✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS ✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS ✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS ✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA ✓ Correct Answer

Answer the questions below

Run the Snort instance and check the build number.

149 ✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1 ✓ Correct Answer

3009 ✓ Correct Answer

Use **snort.log.1640048004**

Read the snort.log file with Snort; what is the IP ID of the 10th packet?

```
snort -r snort.log.1640048004 -n 10
```

49313 ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the referer of the 4th packet?

http://www.ethereal.com/development.html ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the Ack number of the 8th packet?

0x38AFFFF3 ✓ Correct Answer

Read the "**snort.log.1640048004**" file with Snort; what is the number of the **"TCP port 80"** packets?

Room completed: 100%

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170 ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 ✓ Correct Answer

Keep reading the output.How many "HTTP response headers" were extracted?

3 ✓ Correct Answer

Investigate the **mx-1.pcap** file **with the second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

68 ✓ Correct Answer





RESULT: Thus we have learnt how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.

EXPERIMENT NO. 07

SNORT IDS

EXP.NO: 7

```
[root@localhost security]# cd /usr/src

[root@localhost security]# wget https://www.snortIDS.org/downloads/jan/daq-2.0.0.tar.gz

[root@localhost security]# wget https://www.snortIDS.org/downloads/jan/snort-2.0.0.1.tar.gz

[root@localhost security]# tar xvzf daq-2.0.0.tar.gz

[root@localhost security]# tar xvzf snort-2.0.0.1.tar.gz

[root@localhost security]# yum install libpcap* pcre* libdnet* -y

[root@localhost security]# cd daq-2.0.0

[root@localhost security]# ./configure

[root@localhost security]# make

[root@localhost security]# make install

[root@localhost security]# cd jan-2.0.0.1

[root@localhost security]# ./configure

[root@localhost security]# make

[root@localhost security]# make install

[root@localhost security]# jan--version
```

EXPERIMENT NO. 10

MITM ATTACK WITH ETTERCAP

EXP.NO: 10



